



Original research article

Integrating unmanned aerial vehicle-derived habitat metrics with movement persistence and species distribution models to characterize seasonal habitat use of invasive turtles in urban wetlands

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ABSTRACT

Invasive freshwater turtles — the Red-eared Slider (*Trachemys scripta elegans*) and River Cooter (*Pseudemys concinna*) — pose serious ecological threats, yet fine-scale seasonal habitat-use patterns remain poorly understood due to GPS signal loss from semi-aquatic behaviour. We evaluated whether Movement Persistence (MP) modelling, combined with high-resolution unmanned aerial vehicle (UAV) imagery (~2 cm resolution), can identify seasonal habitat suitability and management priority areas for invasive turtles in an urban wetland. Nine individuals were tracked at the Ansan Reed Wetland, South Korea, and season-specific MaxEnt species distribution models (SDMs) were built from MP-reconstructed occurrence proxies. Boosted Regression Trees were additionally fitted to validate variable importance across algorithmically independent approaches. SDMs revealed clear seasonal contrasts: habitat suitability coverage was broader in spring (0.35) than in fall (0.20). Road proximity dominated spring predictions (MaxEnt Percent Contribution: 41.7–54.9%), while non-vegetated islands were the primary fall driver (46.4–53.2%). Both models identified the same leading predictor in the seasonal Total models. Trap deployment guided by SDM outputs achieved 100% spatial overlap with the maximum training sensitivity plus specificity-defined high-suitability zone — covering only 19.2% of the study area (enrichment ratio ≈ 5.2) — yielding over 18 captures in 2022. We propose a seasonally adaptive management strategy and demonstrate an integrated approach combining MP-based trajectory reconstruction, UAV habitat mapping, and multi-model SDMs for invasive chelonian management in urban wetlands.

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1. Introduction

Invasive species management is crucial for maintaining biodiversity, as their introduction often leads to detrimental effects on native ecosystems (Mcneely, 2001). Research focusing on the effective management of these species is essential for devising strategies that mitigate the adverse impacts of non-native species and sustain ecological balance (Mack et al., 2000). Many managers employ strategies like eradication and maintenance control to reduce the spread of invasive fauna (Feuka et al., 2022). In particular, understanding and predicting the movement of invasive species enables the development of more effective containment measures (With, 2002). Such knowledge directly informs management and removal strategies by offering critical insights into how these species traverse and utilize ecosystem environments.

Among invasive species, freshwater turtles such as the Red-eared Slider (*T. scripta elegans*) and the River Cooter (*P. concinna*) significantly contribute to global biodiversity loss through newly introduced predatory and competitive interactions. The Red-eared Slider is included among the world's top 100 most invasive species (Lowe et al., 2000), while the River Cooter has been introduced internationally and is known to inhabit areas similar to those of the Red-eared Slider (Koo et al., 2020, 2023; Pearson et al., 2015). Experimental findings indicate that Red-eared Sliders have a competitive edge over native turtles when competing for food resources (Pearson et al., 2015), while River Cooters can attain larger body sizes, potentially exerting even greater ecological impacts on invaded ecosystems (Koo et al., 2020). Competitive processes arising from trophic niche overlap between invasive and native turtle species have also been reported in natural environments (Balzani et al., 2016). Beyond competitive interactions, the establishment and naturalization of non-native turtles may generate additional ecological pressures through long-term population persistence and expansion within newly colonized habitats (Arizza et al., 2016; Iglesias et al., 2015). Dietary studies further reveal that Red-eared Sliders vary their intake of plant and animal matter seasonally, whereas River Cooters consistently consume filamentous algae (Dreslik, 1999). However, existing research on invasive freshwater turtles has primarily focused on dietary differences, with limited attention given to ecological characteristics related to land type preferences.

Incorporating animal movement knowledge offers valuable perspectives on how invasive species interact with their surroundings, ultimately shaping more effective management plans and ecological studies (Kays et al., 2015; Tabak et al., 2017). Invasive species can cause significant ecological disruptions, yet traditional observation methods often fail to capture fine-scale behaviors critical for targeted control measures. Understanding when and how these animals move is vital for implementing site-specific interventions. Recent studies have advanced beyond merely predicting animal movements to developing indices that capture nuanced behavior patterns. (Jonsen et al., 2023; Joo et al., 2020), thereby helping managers tailor removal strategies. Combining accurate location data with high-resolution spatial information is crucial for effectively managing wildlife, particularly invasive turtle species whose ecological impacts can be severe if left unchecked. However, obtaining reliable location data for semi-aquatic turtles poses a significant challenge. Because these turtles often remain submerged or in habitats with poor GPS reception, traditional monitoring methods frequently produce large data gaps, preventing fine-scale analyses of movement and habitat use. Additionally, research shows that invasive turtles like the Red-eared Slider may avoid areas near roads (Ryan et al., 2008), while certain aquatic vegetation purification zones scarcely limit their dispersal (Angoh et al., 2021). Management priorities often hinge on broader bioclimatic considerations (Ficetola et al., 2008), yet these large-scale approaches may obscure the fine-scale spatiotemporal nuances essential for designing targeted control measures. Therefore, capturing and analyzing granular movement data across seasons, coupled with understanding key environmental drivers, is paramount for refining such strategies (Patterson et al., 2008). In short, the successful management of semi-aquatic invasive turtles demands an integrated analytical approach that not only overcomes data-gathering obstacles but also reveals detailed movement behaviours. By systematically integrating refined location data with high-resolution environmental analysis, researchers and practitioners can more precisely identify critical periods and habitats—ultimately advancing targeted and effective invasive species control.

In parallel with movement-focused analyses, species distribution models provide a powerful tool for predicting animal occurrence under varying environmental conditions (Elith and Leathwick, 2009). By combining high-resolution spatial data with point-level movement insights, researchers can address both where and how invasives move, capturing seasonal habitat shifts that often go undetected in broad-scale assessments. This approach is particularly important for identifying critical times and locations for focused management efforts. Numerous studies demonstrate that ecological and physical factors—such as fine-scale microhabitat features including vegetation-free island distribution, emergent plant patches, floating vegetation cover, and proximity to anthropogenic infrastructure (e.g., roads and boardwalks)—influence the habitat preferences of many species across seasonal phases (e.g., spring active-dispersal and fall pre-overwintering periods) (Bhatnagar et al., 2021; Cardador et al., 2014; Riotte-Lambert and Matthiopoulos, 2020; Ryan et al., 2008; Van Moorter et al., 2013). Incorporating movement-derived occurrence data into these distribution models can further refine predictions of habitat presence (Srivastava et al., 2019). For instance, by examining how environmental parameters drive seasonal movement patterns, researchers have identified specific intervals requiring heightened management (Holloway and Miller, 2017). However, these approaches have rarely been applied at the microhabitat scale necessary for site-specific management of invasive turtles.

Despite these advances, three key gaps remain. First, semi-aquatic species present a fundamental data-collection barrier: extended submergence causes systematic GPS signal loss, producing trajectory datasets too sparse for reliable SDM inputs. Although computational approaches to filling these trajectory gaps have been proposed using AI-based trajectory prediction (Hirakawa et al., 2018) and GPS loggers have been deployed specifically for semi-aquatic freshwater turtles to characterise habitat use and movement corridors (Chow-Fraser, 2014; Ryan et al., 2008), neither approach has been coupled with seasonal SDM modelling to generate spatially explicit management zone candidates for invasive chelonians in urban wetlands. Second, although movement ecology has increasingly been incorporated into SDMs—with systematic reviews noting that finer-scale foraging movements remain the most neglected component

of SDM design (Holloway and Miller, 2017)—most SDM applications to invasive turtles have relied on broad-scale landscape or bioclimatic variables (Ryan et al., 2008; Santoro et al., 2020), overlooking the fine-scale microhabitat features—such as vegetation-free island clusters, emergent plant patches, floating vegetation zones, and proximity to anthropogenic structures including roads and boardwalks—that are likely to govern site-level movement decisions. Third, although state-space models have established the statistical foundation for reconstructing individual movement trajectories from sparse or noisy telemetry data (Patterson et al., 2008), UAV-based drone imagery has demonstrated its capacity to produce fine-scale seasonal vegetation maps of wetland habitats (Bhatnagar et al., 2021), and incorporating temporal constraints across different periods of the annual cycle has been shown to improve SDM predictive performance for mobile species (Cardador et al., 2014), these approaches have not been integrated into a single analytical pipeline. No study has yet combined movement-model-based imputation with UAV-derived fine-scale environmental mapping and SDM to generate season-specific management zone candidates for invasive turtles in urban wetlands. To address these gaps, this study presents an integrated analytical approach that combines GPS telemetry with Movement Persistence (MP) modelling to reconstruct missing location data, UAV-based fine-scale habitat mapping at ~ 2 cm resolution, and seasonal MaxEnt SDMs to better understand the movement and habitat-use patterns of invasive turtles in urban wetlands. To our knowledge, this represents the first study to integrate MP-based trajectory reconstruction with UAV-derived microhabitat covariates and seasonal MaxEnt SDMs for invasive chelonian management in an urban wetland, providing a replicable analytical approach for fine-scale, season-specific management prioritisation.

The primary aim of this study was to evaluate whether MP-enhanced GPS tracking data, combined with UAV-derived fine-scale spatial information, can improve the interpretation of invasive turtle movement and habitat-use patterns in an urban wetland. We hypothesised that (H1) Movement Persistence (MP) modelling could partially reconstruct missing GPS trajectories and provide season-specific occurrence proxies for semi-aquatic invasive turtles, while recognizing that these reconstructed locations are model-derived rather than independent field observations. We further hypothesised that (H2) integrating these MP-derived occurrence proxies with UAV-derived microhabitat variables—including vegetation-free island proximity, emergent plant patch cover, and road distance—in MaxEnt SDMs would reveal season-specific habitat suitability patterns (i.e., spring active-dispersal versus fall pre-overwintering phases) and the environmental drivers underlying each that may inform future monitoring and management prioritisation. To test these hypotheses, this study pursued two objectives: (1) to characterise spatial and seasonal movement patterns of invasive turtles using MP-enhanced tracking data, and (2) to identify potential priority areas through SDM modelling and examine their relevance to field observations using subsequent trapping and recapture records.

2. Materials and methods

2.1. Study site

This study was conducted in Ansan City, Gyeonggi Province, Republic of Korea, specifically at the Ansan Reed Wetland ecological Park ($37^{\circ}19' - 37^{\circ}20'$ N, $126^{\circ}48' - 126^{\circ}49'$ E). This artificial wetland was created in the early 2000s to remove pollutants (See Figs. 1 and 2). It is bordered by an urban area to the north. To the east and south, the wetland is adjacent to agricultural fields, while to the west, it connects to a lake, allowing purified water to flow through. Dominant plant species include *Phragmites communis* (reed), *Typha orientalis* (cattail), *Trapa natans* (water chestnut), and *Nelumbo nucifera* (lotus). Additionally, our study site is home to species classified as Near Threatened (NT) by the IUCN, such as *Lutra lutra* (European otter) (Loy et al., 2022). Despite its artificial origins, the wetland is considered ecologically important and attracts a substantial number of local visitors. However, the prevalence of non-native species highlights the need for ongoing management and conservation efforts to preserve native biodiversity. Although the wetland was originally created as a restoration-oriented ecological park, it continues to be adversely affected by invasive Turtles—particularly Testudines, particularly *T. scripta* (Red-eared Slider) and *P. concinna* (River Cooter)—which pose ongoing threats to native

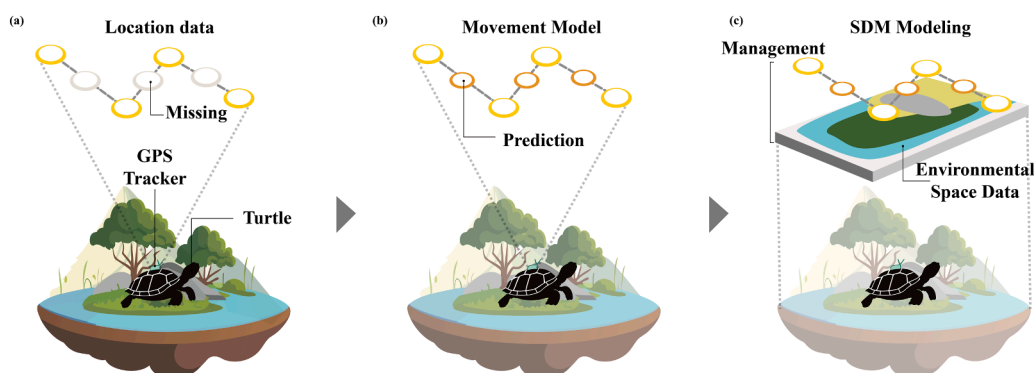


Fig. 1. This figure shows the overall flow of our research. (a) In the field, some turtle location data may be missing for various reasons. (b) We propose a movement-based approach to fill these data gaps. (c) Using the environmental space data, we can build models to predict animal occurrences, leading to more scientific management.

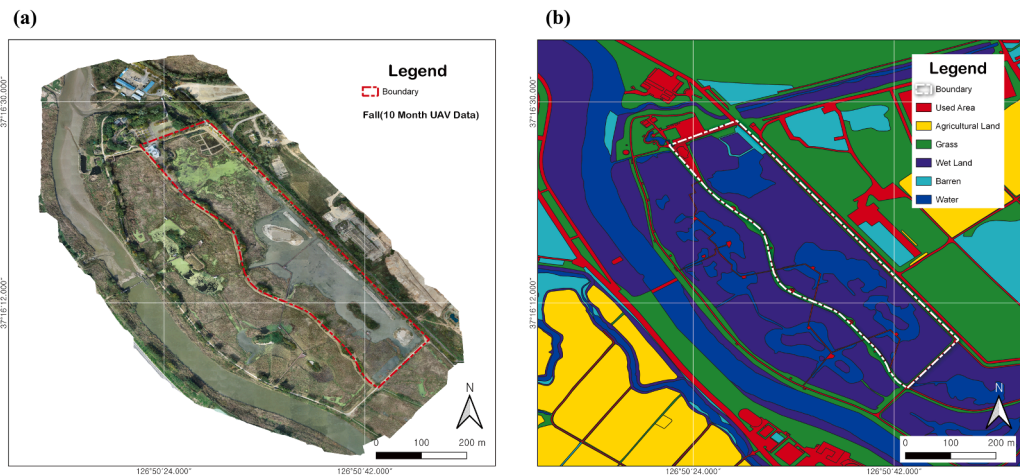


Fig. 2. (a) The study site captured in October 2021 using UAV RGB ortho-mosaic imagery. (b) Land use and land cover (LULC) types derived from coarse 30-m resolution data. The site is primarily composed of wetlands and water, surrounded by a lake, buildings, and grass areas. To address the resolution limitation of the 30-m data, this study re-derived fine-scale cover types by integrating the datasets from (a) and (b).

biodiversity and undermine the ecological functions the site was designed to support (Jung and Lee, 2019; Lowe et al., 2000). Climatically, the region experiences a typical temperate monsoon pattern with hot, humid summers and cold, dry winters, while spring and Fall are relatively mild. These seasonal variations greatly influence the hydrological conditions and ecological processes within the wetland. Despite these climatic fluctuations, invasive turtle species have shown a high capacity for adaptation, enabling them to persist year-round and exert continued pressure on native fauna.

2.2. Target species capture and data collection

In 2020, candidate turtle specimens for this study were provided by field experts responsible for managing the study site. These turtles had been captured in the wild by the site manager using funnel-type traps. Because the turtles were provided from an ongoing site-management capture programme rather than from a trapping campaign designed specifically for this study, the number of traps and trap-nights were not independently quantified as part of the present research design. Instead, the data collection for this study began with the candidate turtles made available by the field manager. Among these candidate individuals, nine adult turtles were selected for GPS tracking based on body size, suitability for tracker attachment, field condition, and management-related release constraints. These nine tracked individuals comprised five *T. scripta* and four *P. concinna*. The tracking experiment was conducted, the required animal ethics training was completed (No. ILAR-01-21-5-010), approval for the animal experiment plan was obtained, and the national capture-release permit was secured (Animal Ethics Experiment Registration No. SNU-220120-2). Because the target non-native turtle species are legally designated as ecosystem-disturbing species in the Republic of Korea, the release of captured individuals required careful consideration to prevent further spread within the study area. At the request of the field manager and relevant governmental institutions, male turtles were excluded from the tracking experiment because their re-release was considered to pose a potential risk of facilitating population expansion. Accordingly, nine adult, predominantly female individuals captured from the field were selected for release, GPS tracking, and subsequent analyses. The tracked species were *T. scripta* (Red-eared slider) and *P. concinna* (River cooter), comprising five individuals of *T. scripta* and four individuals of *P. concinna*. GPS-based location trackers were used (Druid Corporation, LEGO), which achieve approximately 2.5 m positional accuracy under open-sky conditions and provide reliable data for movement reconstruction. The trackers were lightweight, weighing 20 g and measuring $68 \times 21 \times 16$ mm, making them suitable for attachment to small animals such as turtles. To standardize the tracking protocol, the same model of GPS tracker was attached to the same position on the carapace of all individuals following an identical attachment procedure. Five individuals were



Fig. 3. Photograph of invasive freshwater turtles being equipped with GPS-based location trackers and released into the study site.

released and tracked during the fall season of 2021, with GPS locations collected from September to November, whereas four individuals were released and tracked during the spring season of 2022, with GPS locations collected from April to June. Therefore, the tracking dataset represents season-specific movement records collected in different years, rather than repeated tracking of the same individuals across multiple seasons. Although the same tracking protocol was applied to all individuals, the actual GPS fix intervals and data availability varied among individuals because of field conditions, animal behavior, and device-related constraints. Details of the tagging and release procedures, the overall characteristics of the collected location data, and the individual-level tracking duration and GPS fix intervals are provided in [Figs. 3 and 4](#) and [Table 3](#).

The attachment of GPS trackers to the turtles was conducted in compliance with a maximum 5% collar-to-body weight ratio. The GPS trackers were attached to the middle of the turtles' carapaces. To prevent detachment of the sensor device, small notches were made on the marginal scutes along the edge of the carapace, and the device was secured with a string through these notches. No holes were drilled through the vascularized portion of the carapace. This approach was adopted to minimize tissue damage, pain, and potential behavioral alteration. The tracking interval was set to 20–240 min to collect turtle location data. Because turtles utilize both water and land, it was challenging to obtain data when individuals were in deep water (gap area). During the tracking process, the device recharged its battery via solar power. At moments of brief overcharge, it forcibly acquired a large volume of data, resulting in more frequent location data collection. To address this issue, the data were preprocessed. Specifically, location data recorded at intervals of 20 min or less were filtered and adjusted following established procedures ([Tremblay et al., 2006](#)). This filtering procedure was implemented using an open-source package for R-4.40 ([R Core Team, 2024](#)). [Fig. 4](#) illustrates the collection and organization of location data for invasive turtle species. Data collection was possible for approximately two months after release in most cases. One Red-eared slider turtle was monitored continuously for the longest duration, while some individuals could only be tracked for less than a month. The number of data collected per day ranged from a maximum of nine to a minimum of zero, with higher daily collection frequencies observed closer to the release daily. Additionally, as individuals approached the early stages of overwintering, prolonged periods of reduced activity and extended submergence further limited GPS reception, contributing to seasonal gaps in the dataset.

2.3. UAV data collection and land cover type

To collect environmental spatial data for animals with relatively small home ranges, UAV-based imaging was conducted using a DJI Matrice M300 RTK drone ([DJI, 2021](#)). To achieve an approximate ground sampling distance (GSD) of 2 cm, the drone was operated at an altitude of 100 m with a cruising speed of 6 m/s. In flat or low-lying areas, such as agricultural fields or wetlands, an 85% frontal and 70% lateral overlap in imagery is considered sufficient ([Bhatnagar et al., 2021](#)). To minimize positional errors, RTK-based data were collected ([Tomaščík et al., 2019](#)), ultimately generating orthoimage mosaics for October 2021 and April 2022. The RGB orthoimage data were used to redraw fine-scale habitat maps, focusing on landscape features potentially influencing turtle movement and habitat selection.

These UAV-derived maps were supplemented with official land use/land cover (LULC) data provided by government agencies in the Republic of Korea. [Tables 1 and 2](#) summarize this integrative process, wherein categorical LULC data were classified, and distance variables were created to capture the influence of each environmental feature on turtle occurrences. Complementing the coarse-scale land cover data, Fine-scale microhabitat features were incorporated derived from high-resolution UAV imagery, including, including roads, boardwalks, vegetated and non-vegetated islands, emergent vegetation zones, and floating vegetation zones. These features were converted into raster layers at a 3-m resolution to match the GPS precision (~2.5 m). Buffer zones were also created around each feature to assess their potential roles in shaping turtle distributions ([Belkhiria et al., 2016](#); [Santoro et al., 2020](#); [Trebitz et al., 2009](#); [Welicky et al., 2012](#)). A correlation analysis confirmed that all environmental variables had multicollinearity values below 0.7 ([Farrell et al., 2019](#)), ensuring their independent contributions to subsequent modeling processes.

2.4. Model specification: enhanced species distribution modeling for missing data prediction and priority area identification

This section describes a four-step analytical workflow linking GPS-based movement reconstruction, movement persistence modelling, species distribution modelling, and cross-model validation. First, observed GPS locations were processed to summarize individual movement trajectories, tracking duration, movement distance, direction, and home-range extent. Second, movement persistence (MP) modelling was used to reconstruct missing locations caused by irregular GPS reception and to quantify the degree of persistent movement along individual trajectories. Third, the reconstructed movement points were incorporated into the SDM workflow as model-derived occurrence proxies to evaluate how environmental spatial variables influenced seasonal habitat suitability. Fourth, Boosted Regression Trees (BRT) were fitted as a complementary modelling approach to validate the variable importance findings from MaxEnt and to assess whether environmental associations were consistent across algorithmically independent approaches. Thus, the movement reconstruction step addressed data recovery and trajectory characterization, the MP step addressed movement-state quantification, the SDM step addressed habitat suitability and priority-area identification, and the BRT step addressed cross-model robustness of predictor contributions.

The first step was to process observed GPS locations and summarize individual-level movement characteristics. Movement trajectories were processed using the R packages *adehabitatLT* (version 0.3–28) and *adehabitatHR* (version 0.4–22) in R version 4.3.3 ([R Core Team, 2024](#)). These packages are widely used in movement ecology and provide established analytical tools for handling and interpreting animal tracking data ([Joo et al., 2020](#)). The tracking duration for each individual after release was determined as the period during which data were successfully received using these tools. For each individual, Movement-distance trends and final movement direction were examined for each individual. Additionally, using Minimum Convex Polygons (MCP) for behavioral analysis,

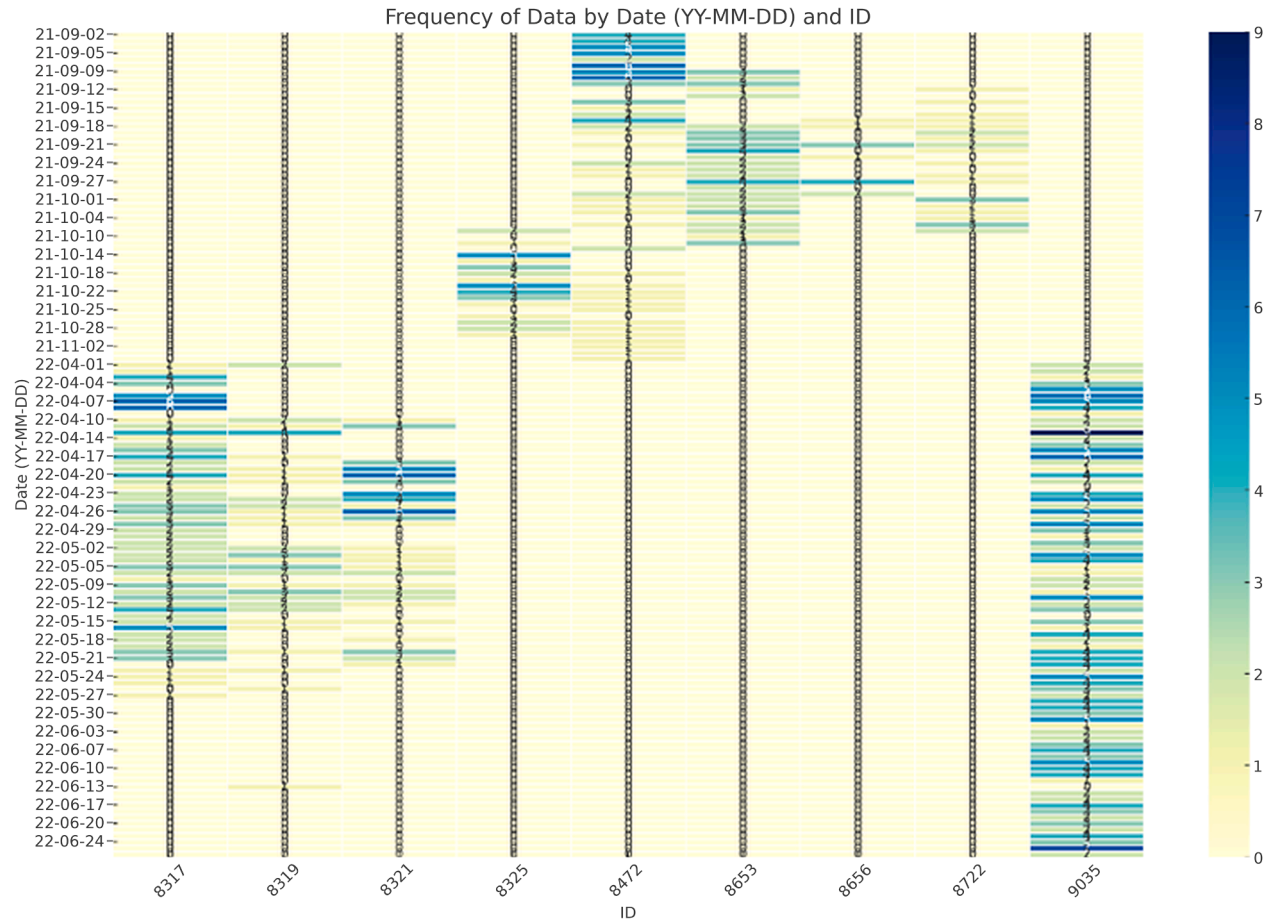


Fig. 4. Location data were obtained for 9 individual invasive turtles. Although data were intended to be collected at 20- to 240-minute intervals, some points were recorded at irregular intervals due to the characteristics of the tracking devices. In addition, extended submergence and reduced activity during transitional periods leading into overwintering resulted in substantial gaps in GPS reception, contributing to further irregularities in the temporal coverage of the dataset.

Table 1

Land use land cover (LULC) classification used for species distribution modeling. The table summarizes the LULC variable name, categorical class contents, data type, and supporting article references based on UAV-collected imagery and supplemental LULC data.

Variables Name	Content	Type	Article Reference(s)
Land Use Land Cover (LULC Type)	1. Open Water Space (OpenW) 2. Vegetation-free Islands (Is-Vf) 3. Vegetated Islands (Is-V) 4. Emergent plant patch (EmP) 5. Grass 6. Road 7. Barren 8. Boardwalk 9. Public facilities 10. Floating plant patch (FleaP)	Categorical	(Belkhiria et al., 2016; Santoro et al., 2020; Trebitz et al., 2009; Welicky et al., 2012)

Table 2

Distance-based environmental variables used to assess the influence of LULC-derived spatial features on turtle occurrences. The table summarizes the buffer-distance variables generated from the LULC classes defined in Table 1, along with the DSM variable, seasonal ranges for spring and fall, data type, and supporting article references.

Variables Name	Content		Type	Article Reference(s)
	Spring	Fall		
Is-Vf_B (LULC: 2, Buffer)	0–350 m	0–350 m	Continuous	(Santoro et al., 2020)
IsV_B (LULC: 3, Buffer)	0–632 m	0–632 m		(Santoro et al., 2020)
EmP_B (LULC: 4, Buffer)	0–124 m	0–125 m		(Trebitz et al., 2009)
FleaP_B (LULC: 10, Buffer)	-	0–137 m		(Trebitz et al., 2009)
OpenW_B (LULC: 1, Buffer)	0–65 m	0–64 m		(Belkhiria et al., 2016)
Boardwalk_B (LULC: 8, Buffer)	0–208 m	0–208 m		(Welicky et al., 2012)
Road_B (LULC: 6; 7; 9, Buffer)	0–97 m	0–97 m		(Santoro et al., 2020)
DSM (Digital Surface Model)	18–45 m	14–42 m		(Welsh and Mansfield, 2022)

Table 3

Using the movement package, we summarized the total number of location points and the start and end dates of the data collection period for each turtle.

Id	Species	Common name	Location points (n. Observation)	Season	Date. Begin	Date. End
8317	<i>Pseudemys concinna</i>	River Cooter	125	Spring	2022–04–01 12:01:01	2022–05–27 10:00:45
8319	<i>Pseudemys concinna</i>	River Cooter	47	Spring	2022–04–01 13:27:44	2022–06–13 12:14:00
8321	<i>Pseudemys concinna</i>	River Cooter	60	Spring	2022–04–11 10:15:37	2022–05–22 12:05:02
8325	<i>Trachemys scripta</i>	Red-eared Slider	34	Fall	2021–10–09 14:16:39	2021–10–29 13:16:19
8472	<i>Trachemys scripta</i>	Red-eared Slider	79	Fall	2021–09–02 11:51:36	2021–11–06 12:16:54
8653	<i>Trachemys scripta</i>	Red-eared Slider	58	Fall	2021–09–09 11:28:48	2021–10–11 14:55:30
8656	<i>Trachemys scripta</i>	Red-eared Slider	12	Fall	2021–09–17 12:50:04	2021–09–30 12:00:54
8722	<i>Pseudemys concinna</i>	River Cooter	25	Fall	2021–09–12 11:05:00	2021–10–09 13:04:38
9035	<i>Trachemys scripta</i>	Red-eared Slider	243	Spring	2022–04–01 12:23:12	2022–06–26 16:23:55

Individual home-range areas were estimated using MCP analysis. The MCP area represents the smallest convex polygon that can enclose each specific area, providing an estimate of the habitat or activity space. The data showcases the diversity in the size of these areas, ranging from less than a hectare to several hectares, indicating variations in the spatial requirements or habitat extents among different regions. Across the nine tracked individuals, 683 observed GPS locations were obtained before movement modelling, with individual observation counts ranging from 12 to 243 points. Data collection in 2021 spanned from September to November, while in 2022, data were collected from April to June (See Table 3). However, as turtles approached periods of reduced activity associated with overwintering, extended submergence and low movement rates resulted in notable declines in GPS reception, limiting data availability during these transitional periods.

The second step was to apply MP modelling to reconstruct missing locations and quantify movement persistence along individual trajectories. Movement analysis was conducted using the Animotum package (Jonsen et al., 2023). This analysis tool allows researchers to quickly derive movement indices by inputting GPS data and related individual information. Movement persistence (MP) analysis was used to broaden the scope of the study by enabling a more detailed examination of turtle movement patterns within the study ecosystem. Model results were then evaluated using AICc. Reconstructed location data were subsequently integrated into the SDM workflow. Because published movement parameters for invasive turtles in wetland systems are limited, The MP model was parameterized directly using movement metrics extracted from our own GPS tracking dataset. The maximum daily displacement (600 m) and maximum movement speed (0.5 m/s) were calculated from the distribution of observed values across all tracked individuals. This approach ensured that the model incorporated biologically realistic thresholds grounded explicitly in field-derived data from this study. For MP-based movement reconstruction, all nine trajectories were initially evaluated. Seven of the nine movement models converged and were retained for subsequent MP-based prediction and SDM analyses, whereas individuals 8317 and 8653 were excluded because their movement models did not converge. The final SDM input therefore consisted of occurrence proxies derived from the seven convergent trajectories, corresponding to 500 observed locations and 661 MP-predicted locations retained for analysis (See Table 4).

The results of the Movement Persistence analysis indicate that out of 9 data sets tracked at 12-hour intervals, 7 successfully converged, meaning that the optimization process was successfully completed and the model identified the best-fitting parameter values for the data, ensuring a stable and well-fitted representation of movement patterns (See Fig. 5). Individuals 8317 and 8653 did not achieve model convergence, likely owing to irregular data collection intervals associated with overcharge-induced burst sampling and extended submergence periods. These individuals were excluded from subsequent SDM analyses to ensure that only reconstructions from convergent MP models were used in the distribution modelling. The analysis yielded AICc (Akaike Information Criterion corrected) values ranging from −104.1 to −1261.6 for each model. Because AICc is a relative measure of model support, its absolute value has no standalone interpretation and may be negative when the maximized log-likelihood is sufficiently large relative to the penalty term. Therefore, the negative AICc values reported here do not indicate model error. In this study, AICc was used to evaluate model support within each individual tracking dataset, rather than to directly compare model fit across individuals, because the number of observed locations and data availability differed among individuals. Despite the limited number of location points, the model generated additional predicted locations, ranging from 27 to 174 points per tracker. "n. Observation" refers to the preprocessed data from each GPS tracker, representing the sample data corresponding to the number of observed locations. Each tracker provided between 12 and a maximum of 243 observed locations for analysis. "n. Prediction" denotes the predicted observation points generated based on "n. Observation" to estimate missing locations. The number of predicted observation points ranged from 27 to 174 per tracker. Among the seven datasets, six were able to supplement location data through the Movement model (See Table 4). To assess reconstruction accuracy, we compared predicted locations against observed GPS fixes occurring within 5 min of a predicted timestamp. Among pairs where model location uncertainty (SE) was below the median ($n = 47$ pairs), the mean positional error was 10.0 m

Table 4

This table presents the predicted results within the Movement Model using location data obtained from GPS trackers. It allows for checking whether the model has converged and comparing the results based on AICc.

Id	Species	Common name	n. Observation	n. Prediction	Converged	AICc
8317	<i>Pseudemys concinna</i>	River Cooter	125	113	FALSE	-972.3
8319	<i>Pseudemys concinna</i>	River Cooter	47	147	TRUE	-332
8321	<i>Pseudemys concinna</i>	River Cooter	60	84	TRUE	-576.3
8325	<i>Trachemys scripta</i>	Red-eared Slider	34	41	TRUE	-228.2
8472	<i>Trachemys scripta</i>	Red-eared Slider	79	132	TRUE	-355.2
8653	<i>Trachemys scripta</i>	Red-eared Slider	58	66	FALSE	-486.6
8656	<i>Trachemys scripta</i>	Red-eared Slider	12	27	TRUE	-104.1
8722	<i>Pseudemys concinna</i>	River Cooter	25	56	TRUE	-187.8
9035	<i>Trachemys scripta</i>	Red-eared Slider	243	174	TRUE	-1261.6

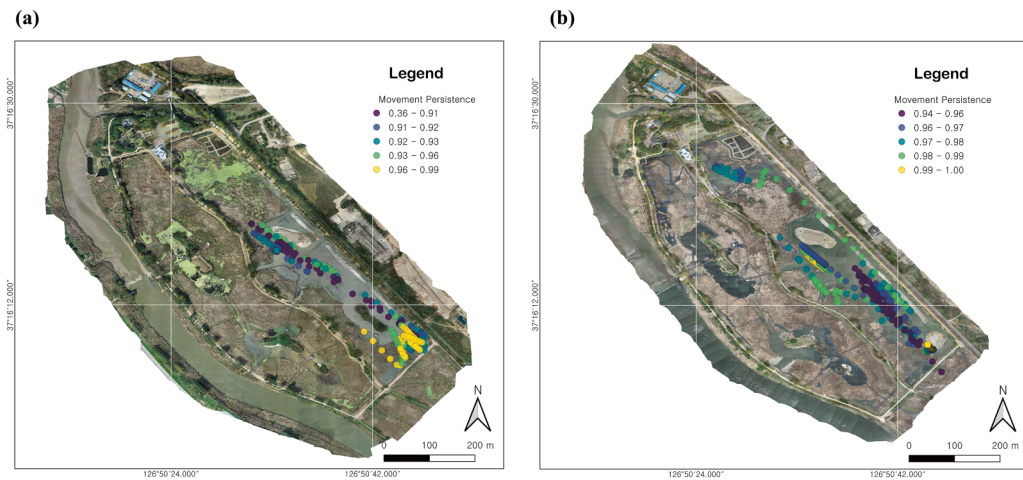


Fig. 5. (a) shows the movement index results based on data acquired during the fall of 2021. (b) shows the movement index results based on data acquired during the spring of 2022.

(RMSE = 17.7 m). This suggests that the continuous-time movement model reliably reconstructed missing locations within biologically meaningful spatial precision.

The third step was to use the reconstructed occurrence proxies in the SDM to evaluate seasonal habitat suitability and identify environmental spatial variables associated with invasive turtle occurrence. Using the predicted observation data derived from "n. Prediction", Changes in predicted distribution were analyzed in the distribution of invasive turtles between the fall of 2021 and the spring of 2022. This analysis was based on average values obtained from the model results across five iterations to assess seasonal variations in species distribution models. Additionally, for each season, Predicted observation data were classified into below-average (values lower than the seasonal mean) and above-average (values higher than the seasonal mean) categories, based on the mean values derived from the results of the Movement Persistence analysis. In the SDM, MP-reconstructed locations were used as occurrence proxies, whereas the above- and below-average MP categories were used only to stratify these occurrence proxies for separate seasonal model comparisons. The predicted locations generated by the MP model were used as model-derived occurrence proxies for the SDM, not as independent field observations. To reduce the risk of unreliable inputs, only predictions from convergent MP models were retained.

To quantify movement-specific characteristics in the reconstructed missing-location data, Seasonal variation in MP values was examined in movement persistence (MP) values derived from the movement model. Specifically, the 2021 dataset was treated as the fall season, whereas the 2022 dataset was treated as the spring season. To maintain data quality, turtles 8317 and 8653 were excluded because their movement models did not converge. The fall dataset, comprising 256 observations, had a mean MP value of 0.93 (SD = 0.06), ranging from 0.35 to 0.99, with Q1, median, and Q3 values of 0.91, 0.91, and 0.95, respectively. In contrast, the spring dataset, comprising 405 observations, showed a higher mean MP value of 0.98 (SD = 0.016), ranging from 0.94 to 1.00, with Q1, median, and Q3 values of 0.96, 0.97, and 0.99, respectively (See Fig. 5). Higher MP values indicate more persistent or directionally consistent movement, whereas lower values suggest more variable, localized, or less persistent movement. Therefore, the higher mean MP value in spring suggests that turtles exhibited more consistent movement trajectories during spring than during fall. This pattern may reflect increased seasonal activity associated with post-winter emergence, foraging, or reproductive movement. However, because the difference between the seasonal mean MP values was relatively small and both seasons showed generally high MP values, this pattern was interpreted as a relative seasonal tendency rather than as conclusive behavioral evidence. This seasonal tendency was then incorporated into the SDM to compare habitat suitability patterns according to relative movement-persistence levels. Therefore, we interpreted this difference as a moderate seasonal tendency rather than evidence of a discrete behavioral shift. By linking these MP-derived occurrence proxies with environmental spatial variables in the SDM, we evaluated how seasonal movement characteristics were associated with habitat suitability and the relative influence of environmental predictors.

The fourth step was to validate the MaxEnt-derived variable importance estimates using Boosted Regression Trees (BRT) as an independent complementary model. Unlike MaxEnt, BRT does not assume a maximum-entropy distribution and derives variable importance from the relative reduction in loss function attributable to each predictor across all trees, providing an independent estimate of predictor contributions that is not contingent on the same algorithmic assumptions as MaxEnt. BRT models were fitted using the `gbm.step` function in the `dismo` package (Elith et al., 2008) in R, with the following hyperparameters: tree complexity = 3 (allowing up to three-way interactions), learning rate = 0.01, and bag fraction = 0.75. Background pseudo-absence points ($n = 10,000$) were randomly sampled from the study area raster extent and used consistently across all datasets. Model complexity (optimal number of trees) was determined automatically via the internal cross-validation procedure in `gbm.step`. Variable importance was quantified as relative influence (RelInf), calculated as the mean percentage contribution to reduction in loss function attributable to each predictor across all trees, averaged over five replicate full-data model fits (each with a different random seed, seed = 43–47) to stabilise importance estimates. Model performance was evaluated using test AUC from five-fold cross-validation for both MaxEnt and BRT,

enabling direct comparison on a common evaluation protocol. For each dataset, occurrence records were partitioned into five folds; MaxEnt used the same folds via its internal cross-validation routine, and BRT withheld each fold sequentially as a test set whilst training on the remaining four folds. Comparisons between MaxEnt Percent Contribution and BRT Relative Influence were made at the level of variable rank order rather than absolute magnitude, as the two metrics are not numerically equivalent — PC is a gain-based metric accumulated during MaxEnt training iterations, whereas RelInf reflects tree-splitting frequency weighted by squared error reduction in BRT.

To assess whether the combined-species occurrence dataset introduced spatial bias in the predicted habitat suitability surfaces, we constructed supplementary species-specific MaxEnt models as a spatial partitioning procedure. The occurrence records comprised two invasive freshwater turtle species — *Pseudemys concinna* and *Trachemys scripta* — whose ecological requirements and movement behaviours may differ despite co-occurring in the same urban wetland system. In this dataset, spring tracking records were predominantly composed of *P. concinna*, whereas autumn records were dominated by *T. scripta*, meaning that species identity and seasonal effect are inherently confounded; without analytical partitioning, taxon-specific spatial habitat signals risk being either structurally masked or entirely lost within the combined suitability surface. To address this spatial imbalance, we partitioned each seasonal occurrence dataset by species identity and constructed separate MaxEnt models for each taxon before combining the resulting suitability maps into a weighted ensemble. Species-specific weights were assigned proportionally to the number of individual animals tracked for each taxon within each seasonal dataset. In spring, three *P. concinna* individuals and one *T. scripta* individual were tracked, yielding weights of 0.75 and 0.25, respectively. In fall, four *T. scripta* individuals and one *P. concinna* individual were tracked, yielding weights of 0.80 and 0.20. These weights were applied exclusively to the predicted suitability maps to generate a spatially adjusted composite visualization; variable importance interpretation throughout this study is based on the combined-species MaxEnt models, as the small number of tracked individuals per taxon in some seasonal groups precludes reliable species-level predictor inference from the partitioned models.

3. Results

3.1. Seasonal distribution analysis of invasive turtles based on location data and movement analysis

During the observation period, daily movement distances ranged from as little as 10 m to as much as 400 m across different individuals. This range indicates that movement activity differed substantially among individuals and seasons, rather than showing a uniform pattern across the entire tracking period. During the spring months (April to June), daily movement distances showed clear individual-level differences. For instance, Turtle (Tag ID: 8317) showed an average movement distance ranging from 17 to 26 m, with a peak of 185.8 m recorded in April. In contrast, Turtle (Tag ID: 8319) displayed higher mobility during spring, with an average movement distance of 62 m in April and a maximum of 344 m. However, by May, this individual's activity had decreased, with an average movement distance of 30 m and a reduced maximum of 145 m. These values indicate that spring movement was not consistently high across all individuals, but was characterized by individual-specific and month-specific variation.

During the fall months (September to November), a contrasting seasonal pattern emerged. In September, movement distances remained relatively high, with some individuals frequently exceeding 150 m in daily maximum distances. However, as October progressed, activity levels declined, with fewer days recording distances above 50 m. By November, movement was significantly reduced, with fewer than 40% of observation days showing distances exceeding 50 m. In addition, more than 50% of the overall observation days had daily movement distances below 50 m as winter approached, indicating a shift toward predominantly short-distance movement.

Overall, the numerical results indicate that spring was associated with higher individual variation and greater mobility, whereas fall was associated with a progressive reduction in movement activity toward winter. Group brumation behavior was not directly observed during the tracking period. Although movement distances declined toward late fall and GPS reception became intermittent during prolonged submergence, these patterns were consistent with reduced seasonal activity but did not provide direct evidence of communal brumation (Liuzzo et al., 2023). However, these movement estimates should be interpreted cautiously because the study site is a semi-anthropogenic wetland with strong physical barriers, including roads, boardwalks, managed channels, and constructed islands. Such barriers may constrain or redirect turtle movement, meaning that the estimated movement distances may reflect both turtle behavior and site-specific landscape structure. Fig. 6 provides a comprehensive visual summary of these seasonal fluctuations, illustrating the contrasting movement patterns between spring and fall across multiple individuals.

The analysis of Minimum Convex Polygon (MCP) percentages provides valuable insights into the spatial dynamics of invasive species (See Fig. 7). Lower MCP thresholds (50–75%) were interpreted as core use areas, representing repeatedly used locations such as daily foraging and basking sites, whereas higher thresholds (85–100%) were interpreted as broader seasonal ranges that also included occasional exploratory or dispersal movements. We focused on thresholds of 85% and above because these levels captured peripheral movement points that were excluded from the core-area estimates but were still ecologically relevant for understanding seasonal habitat use. Among the analyzed MCP values (50–100%), the 90% MCP threshold showed the largest stepwise increase in mean area compared with the preceding threshold, with a mean increment of 6494.89 m² compared to the 85% MCP threshold. Notable examples include ID 8317, where the area expanded from 8405.85 m² at 85% to 23,427.15 m² at 90%, and ID 8321, which showed an increase from 1974.90 m² to 2771.63 m² across the same threshold range. These increases indicate that a small number of peripheral locations substantially expanded the estimated range size. When focusing on the 85–90% MCP range, it became clear that individuals used space differently, as shown by large differences in MCP area. For instance, in ID 8317, the MCP area increased by approximately 2.8 times, illustrating substantial variation in spatial use among individuals. Because MCP estimates were used descriptively to compare spatial-

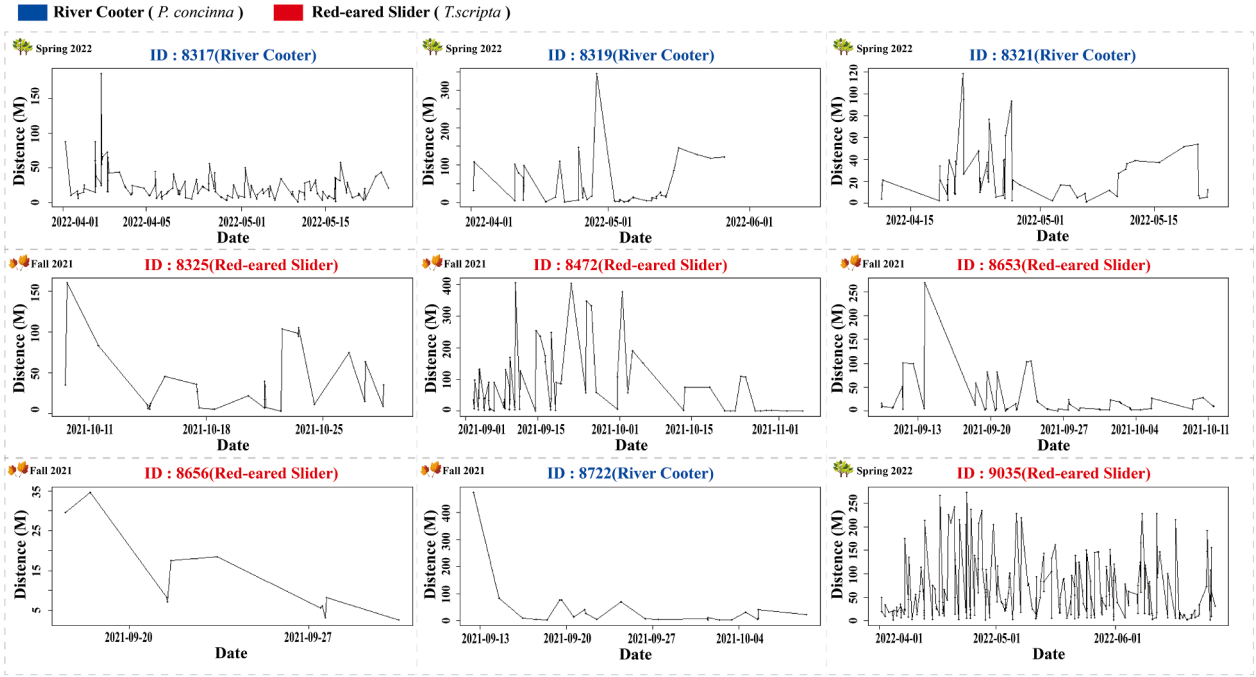


Fig. 6. Time-series of distances between location coordinates for each device (by device ID).

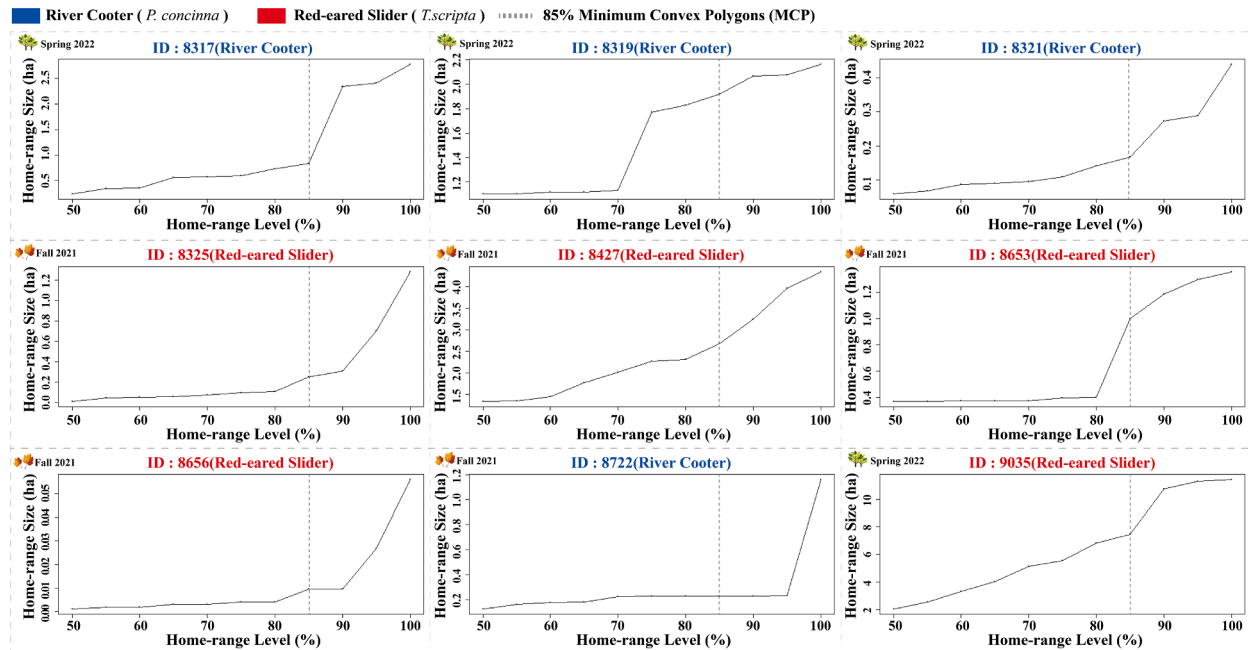


Fig. 7. Minimum Convex Polygon (MCP) areas for various regions (by device ID).

use patterns across thresholds, this increase was interpreted as a numerical expansion of estimated range area rather than as a statistically tested effect.

3.2. SDM results with movement data imputation: exploring seasonal shifts in habitat suitability and land-type drivers

According to SDM predictions, invasive turtle populations exhibited noticeable seasonal differences in habitat suitability and predicted habitat coverage, with the spring model showing higher coverage than the fall model. The mean habitat coverage values across five MaxEnt iterations were 0.355 for spring and 0.202 for fall, corresponding to the rounded values of 0.35 and 0.20 reported here. Habitat coverage was calculated by converting the continuous MaxEnt logistic output into a binary suitable/unsuitable map using the “Equate Entropy of Thresholded and Original Distributions Area” threshold provided by MaxEnt (Fang et al., 2022). This threshold identifies a logistic cut-off at which the entropy of the thresholded prediction is comparable to that of the original continuous suitability distribution. Habitat coverage was then defined as the proportion of grid cells classified as suitable within the modelled area. We used this threshold as a reproducible, model-derived criterion for comparing seasonal habitat coverage, while recognizing that threshold selection is a critical step in presence-only SDMs because continuous suitability outputs often need to be converted into binary maps for ecological interpretation and management applications (Fang et al., 2022; Liu et al., 2013). Because the same modelling procedure, spatial extent, environmental layers, thresholding criterion, and coverage calculation method were applied to both seasonal models, the spring and fall coverage values were considered directly comparable within this study. To account for stochasticity in MaxEnt’s background point selection and regularisation, all models were run across five iterations, and habitat coverage values were averaged across iterations. The coverage estimates were relatively stable across iterations, with coefficients of variation of 6.42% for spring and 3.99% for fall. This threshold equalizes the entropy of the continuous prediction and the binarized output, allowing the derived binary map (suitable/unsuitable) to be used for estimating the spatial extent of suitable habitats. Thus, the coverage values represent the proportion of the study area classified as suitable habitat under this entropy-based threshold. The classification of occurrence proxies based on movement persistence level, above and below the seasonal mean, revealed clearer differences in predicted habitat coverage during spring, whereas such differences were less pronounced in fall. The performance of the models was evaluated using AUC values ranging from 0.85 to 0.93 in Spring and 0.92–0.93 in Fall. These seasonal habitat coverage patterns should be interpreted as model-based indicators of potential priority areas rather than as definitive management prescriptions. We did not conduct an independent targeted trapping experiment specifically designed to validate the SDM predictions. However, during subsequent field trapping activities conducted between May and August 2022, two of the nine tracked individuals were recaptured. These opportunistic recapture records provide supplementary field evidence that some tracked turtles continued to use the monitored wetland system, but they should not be regarded as formal validation of the model outputs. Therefore, the SDM-based coverage estimates are best interpreted as preliminary spatial guidance for future monitoring and targeted trapping, rather than as confirmed management zones.

Importantly, movement persistence (MP Index) in this study was calculated from turtle location data, and individuals were grouped into “Above” and “Below” categories based on the mean MP value. These groups were then used as separate occurrence datasets in MaxEnt modeling to generate three types of SDM predictions for each season: Total models using all imputed occurrence points, Below models using only points from individuals with below-average movement persistence, and Above models using only points from individuals with above-average movement persistence (corresponding to panels a–c for Fall and d–f for Spring in Fig. 8). This grouping structure suggests a more advanced analytical approach by partitioning occurrence data based on movement-derived behavioral indices and explicitly separating models by season, thereby refining both the interpretation of the model and the ecological understanding of animal movement in relation to seasonal habitat dynamics (See Fig. 8 and Table 5). This approach revealed a clear ecological distinction in Spring but not in Fall. In Spring models, turtles in the Below-mean group—representing individuals with lower movement persistence—were associated with higher habitat suitability (average suitability score: 0.20) and broader suitable habitat areas (coverage: 0.34) compared to the Above-mean group (0.17 and 0.19, respectively). This suggests that individuals exhibiting more localized movement during Spring tend to utilize higher-quality or more stable habitats. By contrast, Fall models showed minimal differences between the two groups, with suitability scores (0.15–0.17) and habitat coverage (0.16–0.20) remaining similar, indicating that movement patterns become more homogeneous during this season, potentially due to reduced activity or seasonal environmental convergence. These findings imply that MP-based classification provides meaningful insight into spatial ecology during Spring—useful for identifying priority areas for targeted management—but offers limited additional value in Fall, underscoring the importance of incorporating seasonal behavioral dynamics into SDM interpretation and spatial planning for invasive turtle populations (See Fig. 8 and Table 5).

Beyond differences in habitat suitability, the movement-based grouping approach also revealed substantial seasonal shifts in the environmental drivers shaping invasive turtle distributions. Because the SDM models were separated into Total, Below, and Above categories based on movement persistence, environmental variable contributions could be compared not only across seasons but also across behavioral groups within each season—allowing a more refined interpretation of how movement behavior interacts with landscape features. Advanced SDM analysis of spatial variables showed that key environmental determinants differed markedly between seasons, with road-related variables contributing 41.7–54.9% to habitat suitability in spring, whereas vegetation-free islands were the dominant driver in fall (46.4–53.2%). In spring, the distance from roads (Road_B) was the most influential predictor in the Total model, accounting for 52.02% of the model’s explanatory power (See Table 5). When categorizing the movement persistence (MP) index, this influence became even more pronounced in the below-average group (Spring Season_Below), where road proximity accounted for 54.94% of the model contribution. In contrast, the above-average group (Spring Season_Above) showed a reduced road influence (41.70%) and an increased contribution of vegetation-free islands (Is-Vf_B: 19.97%), indicating a habitat preference shift

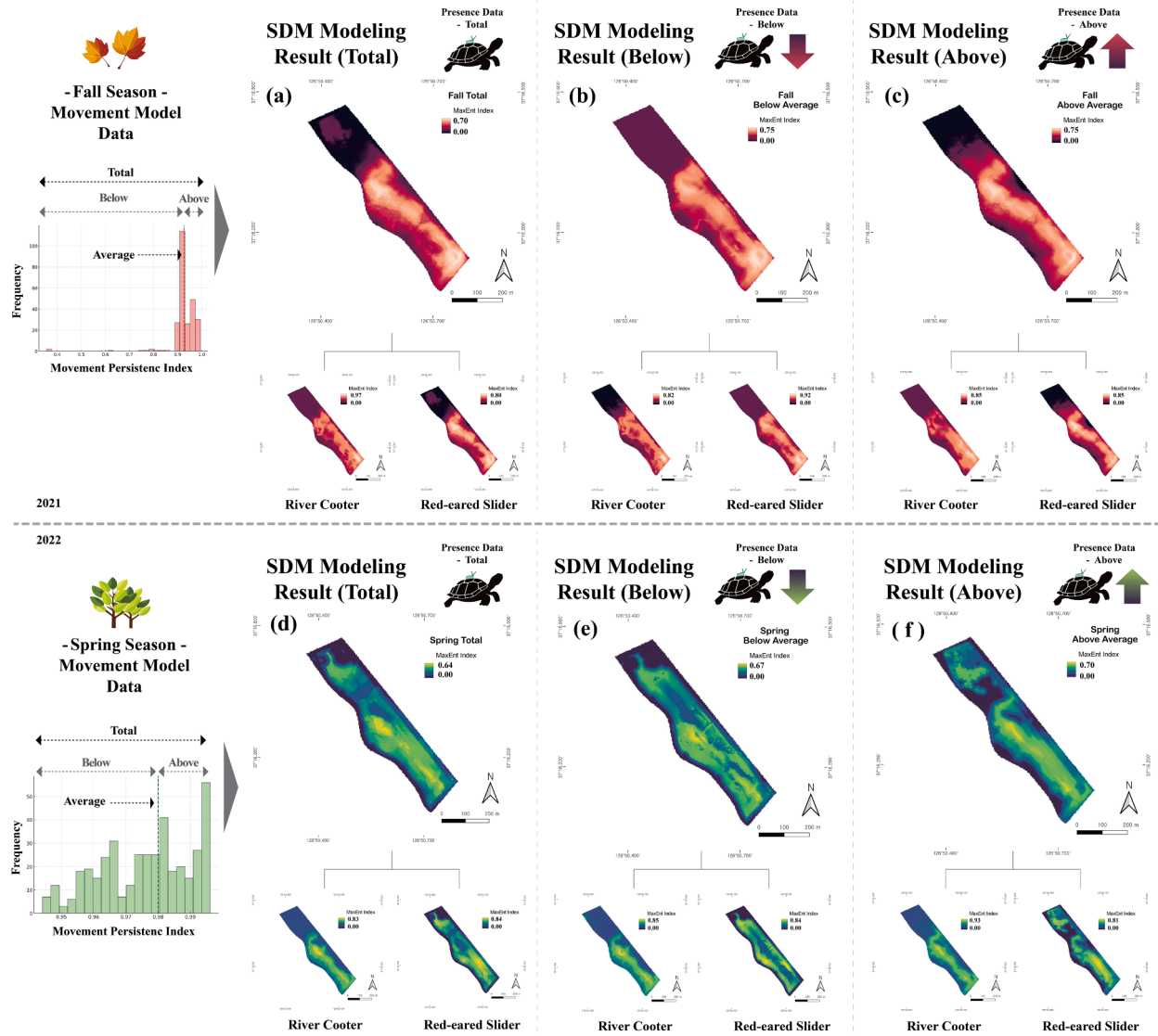


Fig. 8. MaxEnt maps derived from three movement-based occurrence datasets — Total, Below-mean MP, and Above-mean MP — for Fall (a–c) and Spring (d–f), with species-specific predictions weighted by the proportion of tracked individuals per taxon and combined into a seasonal ensemble output.

within the season. In fall, a seasonal transition in key predictors was evident. Roads played a significantly diminished role, explaining only 5.91% of habitat suitability in the total model and dropping to 3.14% in low MP areas. Instead, vegetation-free islands became the dominant factor, contributing 53.17% in the Fall Season (Total) and 51.74% in the Fall Season (Below), consistently representing the primary driver across fall models.

Collectively, these results reveal that the environmental drivers shaping invasive turtle distributions shift in a structured and behavior-dependent manner. In Spring, road proximity dominates overall habitat suitability but becomes even more influential for turtles with localized movement, while individuals with more persistent movement depend less on roads and more on vegetation-free islands. By contrast, Fall models consistently identify vegetation-free islands as the primary driver regardless of movement persistence, with road-related effects nearly disappearing. These behavior–season interactions indicate that habitat selection patterns of invasive turtles cannot be understood by relying on a single season alone; rather, they arise from the combined effects of seasonal environmental conditions and movement-dependent habitat use. Accordingly, effective management strategies must incorporate these empirically demonstrated shifts in environmental drivers.

Both MaxEnt and BRT achieved high discriminative performance across all six seasonal datasets (See Table 5). MaxEnt test AUC values ranged from 0.85 (spring Below-mean) to 0.93 (Fall Above-mean), while BRT AUC values ranged from 0.91 (spring Total) to 0.93 (spring Above-mean). BRT yielded marginally higher AUC values than MaxEnt in spring datasets (Total: 0.91 vs. 0.86; Above-mean: 0.93 vs. 0.93; Below-mean: 0.92 vs. 0.85), whereas the two models performed comparably in Fall across all MP groups. Despite differences in underlying algorithms and importance metrics, MaxEnt and BRT converged on the same primary predictor in the Total model for each season. In spring, Road_B ranked first in both models in the Total and Above-mean datasets (BRT RelInf: 32.5% Total, 49.0% Above-mean), consistent with the MaxEnt PC values reported in Table 5. In Fall, Is-Vf_B was the top-ranked predictor in both Total models (BRT RelInf: 24.6%), again corroborating the MaxEnt results. In the Total models, secondary variable rankings showed some divergence: in spring, BRT elevated Is-V_B (17.5%) and Is-Vf_B (17.4%) above OpenW_B (14.7% in MaxEnt); in Fall, BRT assigned greater importance to EmP_B (23.6% vs. 16.3%) and Boardwalk_B (16.8% vs. 10.8%). These discrepancies likely reflect differences in how each model captures interaction effects rather than genuine uncertainty in variable associations. Taken together, the agreement on the leading predictor in three of the six datasets substantially strengthens confidence that road proximity in spring and non-vegetated island availability in Fall are robust determinants of invasive turtle habitat suitability in this system.

Table 5

AUC values and variable importance scores for MaxEnt and BRT models across six movement-based seasonal datasets. MaxEnt results are presented as Percent Contribution (PC) and BRT results as Relative Influence (RelInf). Environmental spatial variables are described in Tables 1 and 2 of the Methods section.

Input Sample Type(Season)	SDM Model Result (AUC)	Variable And Percent Contribution (%)								
		LULC	Is-Vf_B	Is-V_B	EmP_B	FlaP_B	OpenW_B	Boardwalk_B	Road_B	DSM
Spring Season (Total)	0.86	4.2	12.3	8.8	3.5	-	14.7	2.8	52.0	1.8
Spring Season (Below-mean)	0.85	0.8	6.9	9.9	3.3	-	8.1	8.2	54.9	8.0
Spring Season (Above-mean)	0.93	6.4	20.0	8.1	7.9	-	12.8	3.0	41.7	0.2
Fall Season (Total)	0.92	0.3	53.2	10.2	16.3	1.2	1.6	10.8	5.9	0.6
Fall Season (Below-mean)	0.93	1.3	51.7	17.2	15.9	0.2	0.4	9.4	3.1	0.4
Fall Season (Above-mean)	0.93	1.0	46.4	10.6	12.8	1.1	2.7	10.8	11.2	3.2
Input Sample Type(Season)	BRT Model Result (AUC)	Variable And Relative Influence (%)								
		LULC	Is-Vf_B	Is-V_B	EmP_B	FlaP_B	OpenW_B	Boardwalk_B	Road_B	DSM
Spring Season (Total)	0.91	0.7	17.4	17.5	12.6	-	3.0	9.6	32.5	6.3
Spring Season (Below-mean)	0.92	0.4	17.2	16.7	23.1	-	0.9	15.2	21.1	5.0
Spring Season (Above-mean)	0.93	0.1	12.2	15.9	6.8	-	1.4	6.5	49.0	7.7
Fall Season (Total)	0.92	0.5	24.6	12.7	23.6	7.1	0.6	16.8	10.4	3.3
Fall Season (Below-mean)	0.93	0.1	18.1	14.2	20.1	6.3	0.3	12.2	21.1	7.3
Fall Season (Above-mean)	0.93	0.3	17.4	17.5	32.3	8.3	2.0	8.5	7.5	5.8

4. Discussion

4.1. Management implications

The seasonal patterns identified in this study provide cautious implications for management scheduling. In spring, road-adjacent transitional zones appeared to be strongly associated with predicted habitat suitability, suggesting that monitoring, trap deployment, or barrier-based interventions may be more effective when concentrated near road margins during the active movement period (Ishimoto et al., 2021). In Fall, non-vegetated island clusters—bare-substrate microhabitat features providing high solar exposure suitable for thermoregulation—were more strongly associated with predicted suitability, suggesting that these features may represent potential pre-brumation aggregation sites where monitoring and removal efforts could be prioritised during the pre-overwintering period. Therefore, a seasonally adaptive management strategy, rather than a single annual campaign, may improve the efficiency of invasive turtle control in urban wetland systems, although this interpretation requires evaluation across broader temporal and spatial scales and should account for fine-scale turtle habitat use when implementing habitat management or removal actions (Angoh et al., 2021).

The analytical workflow presented here, which combines MP-based trajectory reconstruction, UAV-derived microhabitat mapping, and season-specific MaxEnt SDMs, offers a transferable analytical approach that may be applicable to other urban or peri-urban wetlands facing comparable chelonian invasion pressures. However, the environmental drivers identified in this study are likely to be site-specific, and direct extrapolation to other systems should proceed with caution pending independent validation. These field results suggest that SDM-derived potential priority areas, when informed by movement persistence classification, may contribute to more targeted trapping or monitoring efforts. However, robust evaluation of efficiency gains over conventional approaches would require replicated field campaigns across multiple seasons and sites.

The present study combined occurrence records from both *T. scripta* and *P. concinna* into a single SDM, reflecting the primary objective of identifying potential priority areas applicable to the invasive chelonian assemblage as a whole. Reliable species-specific inference about predictor importance was not feasible given the limited per-species sample sizes available after GPS preprocessing. In addition, the available tracking data did not show a clear species-level separation sufficient to support robust species-specific habitat models at this spatial scale. Therefore, the combined model was interpreted as representing an assemblage-level invasion signal rather than species-specific habitat selection (Pearson et al., 2007). Another important limitation is that GPS accuracy and reception reliability were likely reduced when turtles were submerged or located in aquatic habitats with poor satellite reception. Although the GPS trackers provide approximately 2.5 m positional accuracy under open-sky conditions, this level of accuracy should not be assumed for periods when individuals were underwater or when signal acquisition was intermittent. These conditions may have contributed to missing or irregular location fixes and should be considered when interpreting the reconstructed trajectories and SDM-derived occurrence proxies. We acknowledge this as a limitation of the present study, and encourage future work with larger, species-specific GPS-tracking datasets to evaluate whether *T. scripta* and *P. concinna* exhibit sufficiently divergent fine-scale habitat affinities to justify independent SDMs and species-tailored management strategies.

4.2. Integrating movement modeling to understand seasonal and spatial dynamics of invasive turtle habitats

The application of the Movement Persistence (MP) index to reconstruct missing GPS trajectories proved ecologically informative, enabling the derivation of season-specific occurrence proxies that would not have been achievable from the raw tracking data alone (Jonsen et al., 2023; Patterson et al., 2008). Critically, the resulting SDM outputs revealed distinct seasonal shifts in habitat-use patterns that have not previously been reported for these two species in an urban wetland context (See Fig. 8). For instance, earlier studies reported that invasive turtles, including the Red-eared Slider, tend to avoid habitats near roads (Ryan et al., 2008) and that certain aquatic vegetation purification zones do not restrict movement (Angoh et al., 2021). However, our seasonal MP index analysis and species distribution modeling revealed that, while distance to roads is a pivotal determinant of turtle distribution in spring, turtles are more likely to be found around non-vegetated islands (Is-Vf_B) in fall. While Horn et al. (2018) incorporated comparable metrics such as basking site availability and surrounding land use, the present study extends this analytical approach through a seasonally explicit analysis, demonstrating that the relative importance of these factors, particularly road proximity and non-vegetated island features, differs markedly between spring and fall (Horn and Gervais, 2018). These findings suggest that the previously noted “road avoidance” behavior (Ryan et al., 2008) may vary by season and, for the first time, elucidate the significant role of island vegetation structure in influencing turtle habitat suitability during Fall.

Additionally, the species distribution models derived from our revised and estimated movement data extend the recommendations of earlier work (Ficetola et al., 2008), which proposed prioritizing invasive species management according to bioclimatic differences, by applying them to more temporally continuous and locally refined spatial scales, specifically discriminating between spring active-dispersal and fall pre-overwintering phases at the microhabitat level. The use of drone-acquired high-resolution data enables the classification of vegetation presence or absence over time and facilitates the identification of specific habitat types, such as emergent vegetation zones, floating vegetation zones, and anthropogenic structures like boardwalks (See Tables 1 and 2). This level of detail allows for the incorporation of fine-scale land type classifications into species distribution models, moving beyond broader features such as roads, water surfaces, or major artificial terrains (See Table 1 and Table 2). This study demonstrates a methodology for revealing seasonally and spatially differentiated patterns of turtle movement and habitat use based on high-resolution, time-series spatial data that cannot be captured through field-based surveys alone (See Fig. 8). However, certain limitations in this study stem from inherent biological characteristics of the focal species and current technological constraints rather than methodological shortcomings.

First, device malfunctions affecting some GPS-tracked turtles shortly after release were likely related to the semi-aquatic lifestyle of the species, including submersion behavior and potential overwintering responses. Although the monitoring period was standardized in design, individual variation in tracking duration was unavoidable. Second, extended submergence periods led to intermittent data loss, reflecting limitations of present-day aquatic sensing capabilities. Because an explicit holdout validation of MP-predicted missing locations was not conducted, the reconstructed locations were interpreted cautiously as model-derived occurrence proxies rather than as directly validated replacements for observed GPS fixes. Third, reconstructed locations should not be regarded as fully independent occurrence records, because their use in SDM may introduce non-independence or potential information leakage. Accordingly, results based on predicted locations were interpreted cautiously, and future studies should include sensitivity analyses using observed GPS locations only or adopt fully integrated movement–distribution models. These data gaps were substantially mitigated through rigorous preprocessing and the implementation of movement modeling approaches, allowing for robust analysis of spatial and behavioral patterns.

A broader methodological consideration concerns the extent to which state-space movement models, including *aniMotum*, account for the physical structure of the study environment when reconstructing animal trajectories. The continuous-time model fitted in this study assumes that movement occurs in an unconstrained space, and does not explicitly encode the hard barriers present in the study wetland — including elevated boardwalks, concrete-lined managed channels, and road embankments — as boundaries to movement. When the model generates predicted locations to supplement observed GPS fixes, these interpolated coordinates may not fully reflect the deflection, detour, or avoidance behaviours that turtles likely exhibit near such anthropogenic structures. In heavily modified urban wetland systems such as the study site, where built infrastructure partitions aquatic habitat into discrete patches, the omission of barrier constraints from the movement model may result in estimated trajectories that underrepresent the true tortuosity of animal paths near structural boundaries.

This limitation has implications for the occurrence proxies used as SDM inputs, particularly for predicted locations generated in the vicinity of boardwalks or channelised margins, where the actual movement path may deviate substantially from the smoothed state-space estimate. Future studies in structurally complex or heavily modified environments would benefit from incorporating barrier-aware movement modelling approaches — for example, by integrating resistance surfaces or employing path re-routing procedures such as those available through the *pathroutr* package — to more faithfully represent constrained movement when generating occurrence proxies for habitat modelling. In the present study, the positional accuracy of the reconstruction was empirically assessed against observed GPS fixes (mean error = 10.0 m; RMSE = 17.7 m; $n = 47$ pairs), providing partial reassurance that predicted locations remained within biologically plausible spatial ranges. Nevertheless, we caution that in environments with high structural complexity, movement model outputs intended for SDM should be interpreted with awareness of the spatial scale at which physical barriers operate, and that integrating barrier geometry into trajectory reconstruction represents an important direction for methodological refinement in urban wildlife tracking studies.

4.3. Linking SDM-derived priority zones with field-based capture outcomes for invasive species management

To validate the utility of the SDM outputs, trapping efforts were conducted in areas identified as high-suitability zones using the maximum training sensitivity plus specificity (MTSS) threshold from MaxEnt, specifically at locations where the above-mean and below-mean MP Index group distributions spatially overlapped (See Fig. 9). To quantify the spatial correspondence between predicted priority zones and trap deployment sites, we calculated a spatial intercept matching percentage: all three trap locations (100%) fell within the MTSS-defined high-suitability zone of the combined spring SDM, which classified only 19.2% of the total study area as suitable — yielding an enrichment ratio of approximately 5.2 relative to random placement. While formal Catch-Per-Unit-Effort (CPUE) metrics were not feasible given the continuous telemetry-based study design, trapping efforts between late July and October 2022 resulted in the capture of over 18 individuals, including *T. scripta* and *P. concinna*, at the designated deployment points. This outcome represents a more efficient capture result within a shorter timeframe compared to previous population surveys conducted at the same site (Jung and Lee, 2019), providing preliminary empirical support for the operational relevance of the SDM-guided priority zones. We acknowledge that a single trapping campaign with three deployment sites is insufficient to draw definitive conclusions; replicated CPUE comparisons within and outside predicted high-suitability strata across multiple seasons would constitute a more rigorous future validation.

5. Conclusion

This study was conducted with two primary objectives: to identify the fine-scale spatial and seasonal movement patterns and distribution of invasive turtle species, and to derive effective management strategies based on these insights. First, we demonstrated that by applying a movement model to semi-aquatic turtles—species prone to frequent GPS signal loss—it is possible to effectively predict seasonal and spatial distribution patterns by compensating for missing tracking data. During the observation period, daily movement distances ranged from as little as 10 m to as much as 400 m across different individuals, demonstrating a substantial degree of variability. In addition, among the analyzed MCP (Minimum Convex Polygon) thresholds ranging from 50% to 100%, broader seasonal ranges were most prominently observed at thresholds of 85% and above, with the 90% MCP threshold showing the most substantial increase in average area—highlighting the influence of peripheral locations. Second, by integrating high-resolution spatial information (e.g., drone-based vegetation and terrain data) into species distribution modeling (SDM), we identified priority management areas and confirmed the practical utility of the model through field-based trapping validation. According to SDM predictions, invasive turtle populations exhibited clear seasonal differences in habitat suitability and coverage. The Spring Season (Total) model

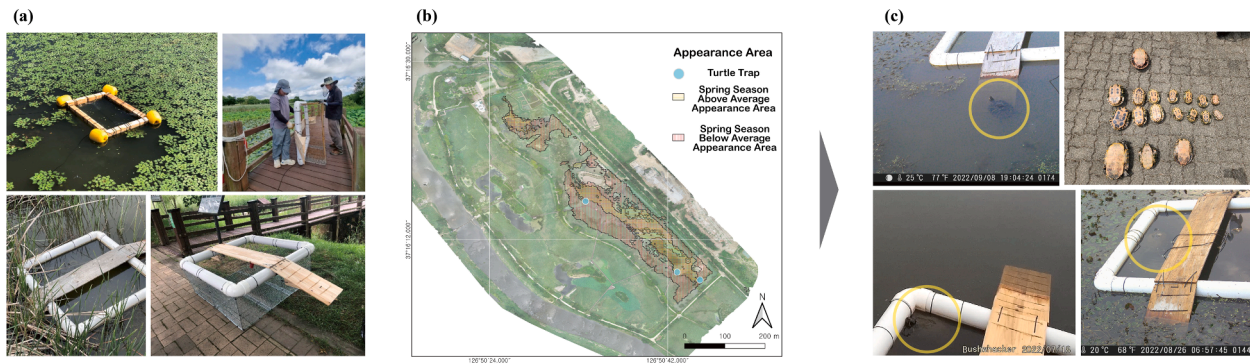


Fig. 9. (a) A basking-exploitation trap was deployed at SDM-predicted high-suitability locations. (b) Placed in predicted appearance areas, the trap successfully captured invasive turtles between July and October, particularly in overlapping distribution zones. (c) The figure displays photographs of the trap and numerous captured individuals at installation sites guided by SDM modeling results.

yielded the highest habitat coverage (0.35), nearly twice that of the Fall Season (Total) model (0.20), indicating more suitable and expansive habitat conditions in spring. Notably, advanced SDM analysis of environmental variables revealed seasonal differences in key determinants, with roads contributing between 41.7% and 54.9% to model performance in spring, and vegetation-free islands contributing between 46.4% and 53.2% in fall. These findings suggest that the previously noted 'road avoidance' behavior may vary by season and, more importantly, highlight finer-scale patterns of habitat use and movement that inform not only seasonal habitat preferences but also effective selection of capture sites.

The significance of this study lies in its ability to address the limitations of GPS data loss by integrating movement models with fine-scale environmental spatial data. This approach enabled the identification of detailed ecological patterns across seasons and habitats for invasive turtles. Furthermore, the methodology can be extended to support empirical monitoring and customized mitigation strategies for a wide range of aquatic and semi-aquatic species. However, the study faced limitations due to a relatively short monitoring period, data variability across individuals and seasons, and occasional malfunctioning of tracking devices. To overcome these constraints, future research should incorporate long-term, all-season monitoring, a larger sample size, and continuous methodological refinement. If implemented, such improvements would not only enhance invasive species management but also broaden the applicability of movement ecology research across diverse taxa.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used [ChatGPT] in order to improve the language and readability of the manuscript (editing and translation). After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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